A TOP-DEGREE PIERI FORMULA FOR DOUBLE GROTHENDIECK POLYNOMIALS

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1. Conventions

Let β be a formal parameter, and write

$$a \ominus b = a + b + \beta ab, \qquad \overline{a} = \frac{-a}{1+\beta a}$$

for the multiplicative formal group law and its formal inverse. Permutations are in one-line notation, $\ell(u)$ denotes the inversion length, and t_{ab} exchanges positions $a < b$. We use the double Grothendieck polynomials $\mathfrak{G}_u(x;y)$ with the normalization

$$\mathfrak{G}_{w_0}(x;y) = \prod_{i+j\le n} (x_i \ominus y_j), \qquad \mathfrak{G}_{us_i}(x;y) = \pi_i^x\, \mathfrak{G}_u(x;y) \quad \text{if } \ell(us_i) < \ell(u),$$

where

$$\pi_i^x f = \frac{(1+\beta x_{i+1})f - (1+\beta x_i)s_i^x f}{x_i - x_{i+1}}.$$

Thus $\mathfrak{G}_{21}(x;y) = x_1 \ominus y_1$.

For an alphabet $\mathcal{Y} = (Y_1, \dots, Y_n)$, write

$$E_{n,n}(\mathcal{Y}; -z) = \prod_{j=1}^{n} (Y_j + z).$$

This is the top-degree factorial elementary symmetric polynomial; in particular,

$$E_{k,k}(x; -z) = \prod_{i=1}^{k} (x_i + z).$$

Definition 1 (Marked k-Pieri support)**.** For $u \in S_\infty$ and $k \ge 1$, let $\mathrm{Supp}_k(u)$ be the set of endpoints of marked k-Pieri chains starting at u. A chain is a saturated increasing Bruhat chain

$$u = u_0 \lessdot u_1 \lessdot \cdots \lessdot u_r = w, \qquad u_j = u_{j-1} t_{a_j b_j}, \quad a_j \le k < b_j,$$

with $b_1 \ge \cdots \ge b_r$, whose covers are marked or unmarked subject to the following conditions:

(1) if $a_j = a_i$ for some $i < j$, then cover j is unmarked;
(2) if $j > 1$ and $\neg\big((b_j < b_{j-1}) \vee (b_j = b_{j-1} \text{ and } a_j > a_{j-1})\big)$, then cover $j-1$ is marked;
(3) every cover in the maximal initial segment with equal b_j and strictly decreasing a_j is marked (in particular, the first cover is marked).

The support consists of endpoints admitting such a marking with at most k marked covers. Equivalently, it is the union, over $0 \le p \le k$, of the supports in the Grothendieck-basis expansions of $e_p(x_1, \dots, x_k)\, \mathfrak{G}_u(x)$.

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2. The formula

For $u, w \in S_\infty$, consider the first k positions of their one-line notations. Define

$$F_k(u,w) = \{i \le k : w(i) = u(i)\},$$
$$L_k(u,w) = \{i \le k : u(i) = w(j) \text{ for some } j < i\},$$
$$O_k(u,w) = \{1,\dots,k\} \setminus \big(F_k(u,w) \cup L_k(u,w)\big).$$

Thus F_k consists of fixed window positions, L_k records values that move left within the window, and O_k consists of values that leave the window or move right within it. Put

$$m_k(u,w) = k - |F_k(u,w)|, \qquad d(u,w) = \ell(w) - \ell(u), \qquad n_k(u,w) = |F_k(u,w)| + |L_k(u,w)|.$$

Define the coefficient prefactor and alphabet

$$\rho_k(u,w) = \beta^{\,d(u,w)-m_k(u,w)}(-\beta)^{|L_k(u,w)|} \prod_{i \in O_k(u,w)} \frac{1}{1+\beta y_{u(i)}},$$
$$\mathcal{Y}_k(u,w) = \big(\overline{y_{u(i)}} : i \in F_k(u,w)\big) \cup \big(-1/\beta\big)^{|L_k(u,w)|}.$$

The order of the entries in $\mathcal{Y}_k(u,w)$ is immaterial.

Theorem 2 (Top-degree double Grothendieck Pieri formula)**.** *For $u \in S_\infty$, $k \ge 1$, and an additional variable z independent of x and y,*

$$\mathfrak{G}_u(x;y)\, E_{k,k}(x;-z) = \sum_{w \in \mathrm{Supp}_k(u)} \rho_k(u,w)\, E_{n_k(u,w),\, n_k(u,w)}\big(\mathcal{Y}_k(u,w);-z\big)\ \mathfrak{G}_w(x;y). \tag{1}$$

The apparent negative powers of β in the alphabet are canceled by the factor $(-\beta)^{|L_k(u,w)|}$ in $\rho_k(u,w)$.

Since $E_{n,n}(\mathcal{Y};-z) = \prod_{Y\in\mathcal{Y}}(Y+z)$, the coefficient of $\mathfrak{G}_w(x;y)$ in (1) can equivalently be written

$$\beta^{\,d(u,w)-m_k(u,w)} \prod_{i \in F_k(u,w)} \left(z - \frac{y_{u(i)}}{1+\beta y_{u(i)}}\right) \prod_{i \in L_k(u,w)} (1-\beta z) \prod_{i \in O_k(u,w)} \frac{1}{1+\beta y_{u(i)}}. \tag{2}$$

The product over an empty set is 1. In particular, this form of the coefficient is regular at $\beta = 0$ on the stated support.

Remark 3 (Classical specialization). At $\beta = 0$, only terms with $d(u,w) = m_k(u,w)$ remain. These are the k-Bruhat Pieri endpoints, and (2) specializes to

$$\prod_{i \in F_k(u,w)} (z - y_{u(i)}).$$

Consequently (1) specializes to the top-degree double Schubert Pieri formula, with the sign of the auxiliary variable determined by $E_{k,k}(x;-z) = \prod_i (x_i + z)$.

Example 4. Take $u = 21$ and $k = 2$. The support is $\{21, 231, 312, 321\}$, and the formula gives

$$\begin{aligned}(x_1+z)(x_2+z)\,\mathfrak{G}_{21}(x;y) = {}& (z+\overline{y_2})(z+\overline{y_1})\,\mathfrak{G}_{21}(x;y) \\ & + \frac{z+\overline{y_2}}{1+\beta y_1}\,\mathfrak{G}_{231}(x;y) + \frac{z+\overline{y_1}}{1+\beta y_2}\,\mathfrak{G}_{312}(x;y) \\ & + \frac{1}{(1+\beta y_1)(1+\beta y_2)}\,\mathfrak{G}_{321}(x;y).\end{aligned}$$